Companion LVII: Weak-Lensing Tomography in the Relaxation-Driven Cyclic Cosmology

Michael Lehmann

mi.lehmann@gmx.de

April 2026

Abstract

Weak-lensing tomography probes the redshift evolution of the gravitational potential by correlating shear fields in multiple source bins. In the standard cosmological model, the tomographic kernels are determined by the matter power spectrum, the growth rate, and the geometry of the Universe. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the lensing kernels, the growth of structure, and the redshift dependence of the shear correlations in a characteristic way. This Companion develops a continuous description of weak-lensing tomography in RDCC, deriving how the relaxation scale influences the tomographic kernels, the cross-bin correlations, and the observational signatures in cosmic shear surveys. The resulting framework provides a distinctive probe of the RDCC gravitational field across cosmic time.

1 Introduction

Weak gravitational lensing probes the integrated gravitational potential along the line of sight. Tomographic lensing extends this by dividing source galaxies into redshift bins, allowing the reconstruction of the redshift evolution of the gravitational field. The tomographic kernels encode the geometry of the Universe, the matter distribution, and the growth of structure.

In the standard cosmological model, the tomographic kernels evolve smoothly with redshift. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the tomographic kernels, the growth rate, and the cross-bin correlations in a scale-dependent manner, producing a distinctive signature in weak-lensing tomography.

2 The RDCC Lensing Kernel

The standard lensing kernel for a source bin i is

$$W_i(\chi) = \frac{3H_0^2\Omega_m}{2c^2}\frac{\chi}{a(\chi)}\int_\chi^{\chi_H} d\chi'\, n_i(\chi')\,\frac{\chi'-\chi}{\chi'}. \tag{1}$$

In RDCC, the gravitational potential evolves as

$$\Phi_{\mathrm{RDCC}}(k,a) = \Phi(k,a)\left[1-\chi_\Phi\left(\frac{k}{k_{\mathrm{rel}}}\right)^\alpha\right]. \tag{2}$$

The RDCC lensing kernel becomes

$$W_{i,\mathrm{RDCC}}(\chi) = W_i(\chi)\left[1-\chi_W\left(\frac{k}{k_{\mathrm{rel}}}\right)^\alpha\right]. \tag{3}$$

3 Tomographic Power Spectra in RDCC

The tomographic shear power spectrum between bins i and j is

$$C_\ell^{ij} = \int_0^{\chi_H} d\chi \, \frac{W_i(\chi)\, W_j(\chi)}{\chi^2} \, P_m \left(k = \frac{\ell}{\chi}, z(\chi) \right). \tag{4}$$

In RDCC, the matter power spectrum becomes

$$P_{m,\mathrm{RDCC}}(k,z) = P_m(k,z) \left[1 - \chi_P \left(\frac{k}{k_{\mathrm{rel}}} \right)^\alpha \right]. \tag{5}$$

The tomographic spectrum becomes

$$C_{\ell,\mathrm{RDCC}}^{ij} = C_\ell^{ij} \left[1 - \chi_C \left(\frac{\ell}{\ell_{\mathrm{rel}}} \right)^\alpha \right]. \tag{6}$$

4 Cross-Bin Correlations and RDCC Signatures

The relaxation mechanism enhances the coherence of large-scale modes. The cross-bin correlations therefore exhibit:

- enhanced correlations at large angular scales,
- suppressed correlations at small scales,
- a characteristic turnover near ℓ_{rel},
- a modified redshift dependence of the cross-bin amplitude.

These signatures provide a direct observational probe of the RDCC gravitational field across cosmic time.

5 Redshift Evolution of the RDCC Kernel

The redshift evolution of the RDCC kernel is given by

$$\frac{\partial W_{i,\mathrm{RDCC}}}{\partial z} = \frac{\partial W_i}{\partial z} \left[1 - \chi_W \left(\frac{k}{k_{\mathrm{rel}}} \right)^\alpha \right] - W_i \, \chi_W \, \alpha \left(\frac{k}{k_{\mathrm{rel}}} \right)^\alpha \frac{\partial \ln k_{\mathrm{rel}}}{\partial z}. \tag{7}$$

This term is unique to RDCC and provides a direct probe of the relaxation scale.

6 Conclusion

Weak-lensing tomography in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential and the matter distribution. The relaxation mechanism modifies the tomographic kernels, the growth rate, and the cross-bin correlations in a scale-dependent manner, producing distinctive signatures in cosmic shear surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of cosmic shear, matter evolution, and the interplay between light and gravity in RDCC.

References

[Bartelmann and Schneider(2001)] M. Bartelmann and P. Schneider, Physics Reports **340**, 291 (2001).

[Hu(1999)] W. Hu, The Astrophysical Journal **522**, L21 (1999).

[Hu(2002)] W. Hu, Physical Review D **65**, 023003 (2002).

[Takada and Jain(2004)] M. Takada and B. Jain, Monthly Notices of the Royal Astronomical Society **348**, 897 (2004).

[Kilbinger(2015)] M. Kilbinger, Reports on Progress in Physics **78**, 086901 (2015).

[Collaboration(2022)] D. Collaboration, Physical Review D **105**, 023520 (2022).

[Collaboration(2021)] K. Collaboration, Astronomy & Astrophysics **646**, A140 (2021).

[Collaboration(2023)] H. Collaboration, Publications of the Astronomical Society of Japan **75**, 1 (2023).

[Joachimi and Bridle(2010)] B. Joachimi and S. Bridle, Astronomy & Astrophysics **523**, A1 (2010).